Differential Dynamic Programming Formulations for Multi-Period Optimal Power Flow

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I. Problem Formulations

A. Copper-Plate MPOPF

A storage device and grid connection serve demand p_L^t over $\mathcal{T} = \{1, \ldots, T\}$. Here $P_B^t > 0$ discharges, and B^t is the end-of-period energy; hence B^0 is initial and B^T is terminal. The sign convention follows [1].

$$\min_{P_{\text{Subs}},\, P_B,\, B} \sum_{t=1}^{T} \Big[c^t P_{\text{Subs}}^t + C_B \big(P_B^t\big)^2 \Big] \Delta t + w\big(B^T - B_{\text{init}}\big)^2 \tag{1a}$$

$$\text{s.t.} \quad P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \qquad t \in \mathcal{T} \tag{1b}$$

$$B^0 = B_{\text{init}} \tag{1c}$$

$$B^t = B^{t-1} - P_B^t \Delta t, \qquad t \in \mathcal{T} \tag{1d}$$

$$P_{\text{Subs}}^t \geq 0, \qquad t \in \mathcal{T} \tag{1e}$$

$$\underline{P_B} \leq P_B^t \leq \overline{P_B}, \qquad t \in \mathcal{T} \tag{1f}$$

$$\underline{B} \leq B^t \leq \overline{B}, \qquad t \in \mathcal{T} \tag{1g}$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. SOCP MPOPF

Let $\mathcal{N}$, $\mathcal{L}$, $\mathcal{B}$, and $\mathcal{D}$ denote buses, directed branches, battery buses, and PV buses; $j_{\text{subs}} = 1$ is the substation bus and $\mathcal{C}(j)$ contains the children of j. The network MPOPF is

$$\min \sum_{t\in\mathcal{T}} c^t S_{\text{base}} P_{\text{Subs}}^t \Delta t + \sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{B}} C_B \big(S_{\text{base}} P_{B,j}^t\big)^2 \Delta t + w \sum_{j\in\mathcal{B}} \big(B_j^T - B_{\text{init},j}\big)^2 \tag{2}$$

$$\text{s.t.} \quad P_{\text{Subs}}^t - \sum_{(j_{\text{subs}},k)\in\mathcal{L}} P_{j_{\text{subs}}k}^t = 0, \tag{3}$$

$$P_{\text{Subs}}^t \geq 0, \tag{4}$$

$$Q_{\text{Subs}}^t - \sum_{(j_{\text{subs}},k)\in\mathcal{L}} Q_{j_{\text{subs}}k}^t = 0, \tag{5}$$

$$\sum_{k\in\mathcal{C}(j)} P_{jk}^t - P_{ij}^t + r_{ij}\ell_{ij}^t = P_{B,j}^t + p_{D,j}^t - p_{L,j}^t, \tag{6}$$

$$\sum_{k\in\mathcal{C}(j)} Q_{jk}^t - Q_{ij}^t + x_{ij}\ell_{ij}^t = q_{D,j}^t - q_{L,j}^t, \tag{7}$$

$$v_j^t = v_i^t - 2(r_{ij}P_{ij}^t + x_{ij}Q_{ij}^t) + (r_{ij}^2 + x_{ij}^2)\ell_{ij}^t, \tag{8}$$

$$(P_{ij}^t)^2 + (Q_{ij}^t)^2 \leq v_i^t \ell_{ij}^t, \tag{9}$$

$$\ell_{ij}^t \geq 0, \tag{10}$$

$$v_{\text{Subs}}^t = 1.05^2, \tag{11}$$

$$\underline{V}_j^2 \leq v_j^t \leq \overline{V}_j^2, \tag{12}$$

$$-q_{D,j}^{\max,t} \leq q_{D,j}^t \leq q_{D,j}^{\max,t}, \tag{13}$$

$$q_{D,j}^{\max,t} = \sqrt{S_{D,j}^2 - (p_{D,j}^t)^2}, \tag{14}$$

$$B_j^t = B_j^{t-1} - P_{B,j}^t \Delta t, \tag{15}$$

$$B_j^0 = B_{\text{init},j}, \tag{16}$$

$$\text{soc}_j^{\min} B_{R,j} \leq B_j^t \leq \text{soc}_j^{\max} B_{R,j}, \tag{17}$$

$$-P_{B,j}^R \leq P_{B,j}^t \leq P_{B,j}^R. \tag{18}$$

II. Simulation Setup

TABLE I
Copper-Plate Instance Data

Interval t	1	2	3	4	5	6
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140

Period length Δt	1	Initial energy B_{init}	2.0
Throughput C_B	0.05	Terminal weight w	5.0

Bounds, constrained variants only	
$\underline{P_B}$, $\overline{P_B}$	-0.45, 0.45
$\underline{B}$, $\overline{B}$	1.20, 1.95

All quantities are in per unit on a 1000-kVA base, 1 p.u. $=$ 1000 kW and 1 p.u.h $=$ 1000 kWh. Bounds apply only to the constrained variants.

III. Implementation and Results

The network transcription uses analytic derivatives and accepts time-varying load, PV, and price data directly at each stage. Compact problems retain FilterDDP's dense null-space backward pass. For feeder models, sparse Jacobians and local Hessian blocks are instead assembled into the saddle-point system

$$\begin{bmatrix} H & A^\mathsf{T} \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta\lambda \end{bmatrix} = b, \tag{19}$$

which is solved by sparse LU without explicitly constructing a dense QR basis or reduced Hessian. The dense route has less overhead for small problems; the sparse route is required when those dense intermediate matrices no longer fit practical memory. Reported FilterDDP times include its optimization iterations, not model construction. Centralized references use the exact native second-order cone equivalent of (9) in Gurobi and were checked by an independent constraint validator.

A. IEEE123C

The loss-aware model was first tested on IEEE123C (128 buses, 127 branches, 51 batteries, and 51 PV units), with per-stage dimensions $(n_x, n_u, n_c) = (51, 791, 562)$. FilterDDP returned normally for $T \in \{3, 6, 12, 24, 48, 96\}$; its largest equality residual was 8.9×10^{-8} and its largest absolute objective gap from the centralized reference was 5.5×10^{-4}. Runtime increased from 74.9 s at $T = 3$ to 2021.4 s at $T = 96$ (48 and 121 iterations, respectively). The corresponding centralized solve required 12.57 s at $T = 96$. This establishes functional horizon scaling, but not competitive runtime.

B. Corrected IEEE2522C

The feeder previously labelled "IEEE2552C" was renamed IEEE2522C. Its one-phase aggregate conversion had also combined aggregate three-phase powers with a phase-voltage base, making per-unit line impedances nine times too large. Only the electrical base was corrected, to 7.2 kV on 1 MVA; physical impedances, powers, DER ratings, voltage limits, losses, and the SOC model were unchanged. With this correction, cold-start centralized models were globally optimal and independently validated through $T = 96$.

Here $(n_x, n_u, n_c) = (250, 13358, 10337)$ per stage. The original dense backward pass could not complete one iteration because it attempted a dense 13358×13358 QR basis and a reduced Hessian of order 3021. The sparse KKT route removes that storage bottleneck and gives the verified results in Table II. All runs used full line-loss balances and the exact SOC constraint, without warm starts.

TABLE II
SPARSE FILTERDDP ON CORRECTED IEEE2522C

T	Iter.	Time (s)	$\lvert J - J^\star \rvert$	Eq. residual
3	56	233.098	7.34×10^{-4}	5.63×10^{-10}
6	82	836.933	1.30×10^{-3}	1.75×10^{-9}
12	79	1902.103	9.21×10^{-5}	2.29×10^{-8}
24	84	3068.228	1.22×10^{-3}	3.61×10^{-8}

Thus the method is numerically and memory feasible at a 24-period horizon on this feeder, but remains slow: the $T = 24$ run took 51.1 min, versus 27.2 s of Gurobi solver time for the centralized reference. Reusing UMFPACK's symbolic factorization did not help: the identical $T = 3$ solution slowed from 233.1 to 295.3 s. Better scaling therefore requires a more suitable numerical factorization or exploitation of feeder structure, rather than symbolic-cache reuse alone.

C. Large10kC Boundary

The synthetic large10kC feeder also required a data correction before any OPF comparison was meaningful. Its aggregate voltage base was restored to 12.47 kV, and uniform synthetic ordinary-line sections were shortened from $(r, x) = (0.07, 0.01)\ \Omega$ to $(0.0175, 0.0025)\ \Omega$; area-boundary sections use $(0.000025, 0.000025)\ \Omega$. Powers, DER capacities, voltage limits, loss terms, and SOC constraints were retained. Cold-start Gurobi solutions then validated at $T = 1$ and $T = 3$; the latter has 133035 variables, objective 2985281.3189, 27 barrier iterations, and 6.304 s solver time.

For FilterDDP at $T = 3$, the per-stage dimensions are $(n_x, n_u, n_c) = (1020, 54665, 42303)$. Blocking the multiple right-hand-side KKT solve bounded memory and avoided the earlier immediate allocation failure. However, the run produced no terminal iteration report after two hours and was stopped manually. Consequently, no FilterDDP convergence or timing claim is made for large10kC. The experiment instead identifies sparse numerical factorization and feedback-solve cost as the present scalability boundary.

APPENDIX A
FIRST-ORDER DDP DERIVATION

This appendix states the first-order Differential Dynamic Programming workflow directly in MPOPF notation. Equality multipliers are denoted by λ and inequality multipliers by μ. Thus the dynamics multiplier called $\mu[t]$ in the earlier derivation is written λ^t_{soc} here.

A. Centralized Lagrangian

Write every inequality as $g \le 0$. The Lagrangian of (1) is

$$\begin{aligned}\mathcal{L} = &\sum_{t=1}^{T}\left[c^t P^t_{\text{Subs}} + C_B (P^t_B)^2\right]\Delta t + w(B^T - B_{\text{init}})^2\\ &+ \sum_{t=1}^{T} \lambda^t_{\text{bal}}(P^t_{\text{Subs}} + P^t_B - p^t_L)\\ &+ \sum_{t=1}^{T} \lambda^t_{\text{soc}}(B^t - B^{t-1} + \Delta t P^t_B)\\ &+ \sum_{t=1}^{T} \mu^t_{\underline{P_{\text{Subs}}}}(-P^t_{\text{Subs}})\\ &+ \sum_{t=1}^{T}\left[\mu^t_{\underline{P_B}}(\underline{P_B} - P^t_B) + \mu^t_{\overline{P_B}}(P^t_B - \overline{P_B})\right]\\ &+ \sum_{t=1}^{T}\left[\mu^t_{\underline{B}}(\underline{B} - B^t) + \mu^t_{\overline{B}}(B^t - \overline{B})\right]. \end{aligned} \tag{20}$$

Four optimality conditions. Here $B^0 = B_{\text{init}}$ is fixed. In addition to primal feasibility, dual feasibility $\mu \ge 0$, and complementary slackness, stationarity gives four equations. The first two hold for $t = 1, \ldots, T$, the third for $t = 1, \ldots, T-1$, and the fourth at T:

$$0 = c^t \Delta t + \lambda^t_{\text{bal}} - \mu^t_{\underline{P_{\text{Subs}}}}, \tag{21}$$

$$\begin{aligned}0 = &\, 2C_B P^t_B \Delta t + \lambda^t_{\text{bal}} + \Delta t \lambda^t_{\text{soc}}\\ &- \mu^t_{\underline{P_B}} + \mu^t_{\overline{P_B}},\end{aligned} \tag{22}$$

$$0 = \lambda^t_{\text{soc}} - \lambda^{t+1}_{\text{soc}} - \mu^t_{\underline{B}} + \mu^t_{\overline{B}}, \tag{23}$$

$$0 = 2w(B^T - B_{\text{init}}) + \lambda^T_{\text{soc}} - \mu^T_{\underline{B}} + \mu^T_{\overline{B}}. \tag{24}$$

Equations (21) and (22) are the power stationarity conditions visible inside a period. Equations (23) and (24) make those individually optimal periods multi-period optimal by chaining the marginal value of energy through time and terminating that chain correctly. All four residuals were checked against the centralized JuMP/Ipopt solution; the largest was 2.2×10^{-15} for $T \in \{3, 6\}$.

B. Per-Period Decomposition

At sweep k, period t receives $\boldsymbol{B^{k,t-1}}$ from the period just solved and the successor multiplier $\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}}$ from the previous sweep. It solves

$$\min_{P_{\mathrm{Subs}}^t, P_B^t, B^t} \quad c^t P_{\mathrm{Subs}}^t \Delta t + C_B (P_B^t)^2 \Delta t - \boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t \tag{25}$$

$$\text{s.t.} \quad P_{\mathrm{Subs}}^t + P_B^t = p_L^t, \tag{26}$$

$$B^t - \boldsymbol{B^{k,t-1}} + \Delta t P_B^t = 0, \tag{27}$$

$$P_{\mathrm{Subs}}^t \geq 0, \quad \underline{P_B} \leq P_B^t \leq \overline{P_B}, \quad \underline{B} \leq B^t \leq \overline{B}. \tag{28}$$

For $t = T$, the last term of (25) is replaced by $w(B^T - B_{\mathrm{init}})^2$. After solving, the equality multiplier is read from the local SOC equation,

$$\lambda_{\mathrm{soc}}^{k,t} = -\,\mathrm{dual}\big(B^t - \boldsymbol{B^{k,t-1}} + \Delta t P_B^t = 0\big)\,. \tag{29}$$

The term $-\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t$ inserts the successor term missing from a standalone OPF's stationarity in B^t. The earlier program wrote the equivalent expression

$$\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} \left(B^{k-1,t+1} - B^t + \Delta t P_B^{t+1}\right). \tag{30}$$

Because $B^{k-1,t+1}$ is fixed and the auxiliary P_B^{t+1} is not the next period's actual decision, only the $-\boldsymbol{\lambda_{\mathrm{soc}}^{k-1,t+1}} B^t$ part supplies the intended inter-period sensitivity.

C. Sweep

The workflow is

$$t = 1 \to T: \quad \boldsymbol{B^{k,t-1}} \to (25) \to (B^{k,t}, \lambda_{\mathrm{soc}}^{k,t}), \qquad k \leftarrow k{+}1, \tag{31}$$

and stops when

$$\max\big(\|B^k - B^{k-1}\|_2, \|\lambda_{\mathrm{soc}}^k - \lambda_{\mathrm{soc}}^{k-1}\|_2\big) < \epsilon. \tag{32}$$

This passes only the first derivative of the future cost. Filter-DDP also passes its curvature V_{BB}^t; the first-order workflow has no corresponding quantity.

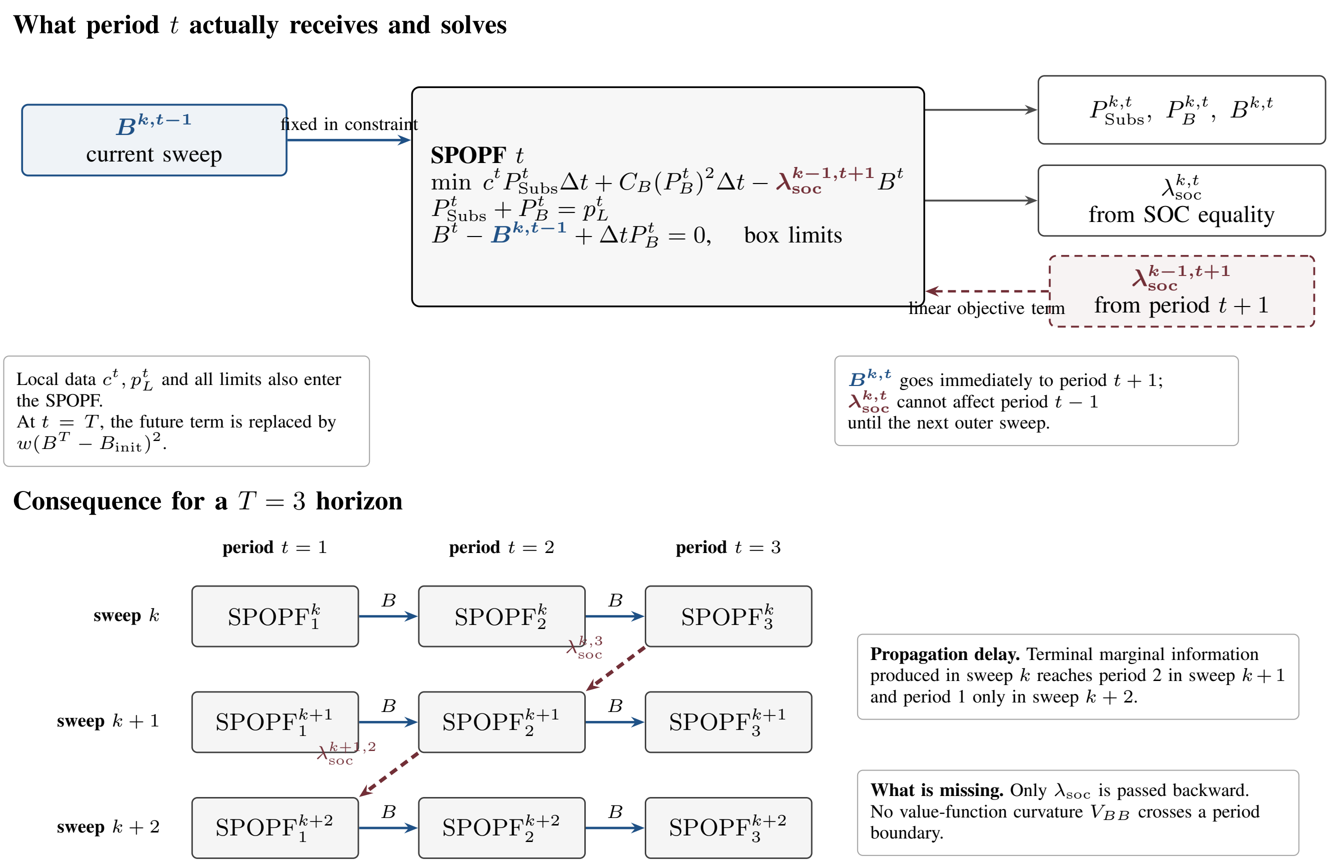

Fig. 1. Information flow in the first-order DDP sweep.

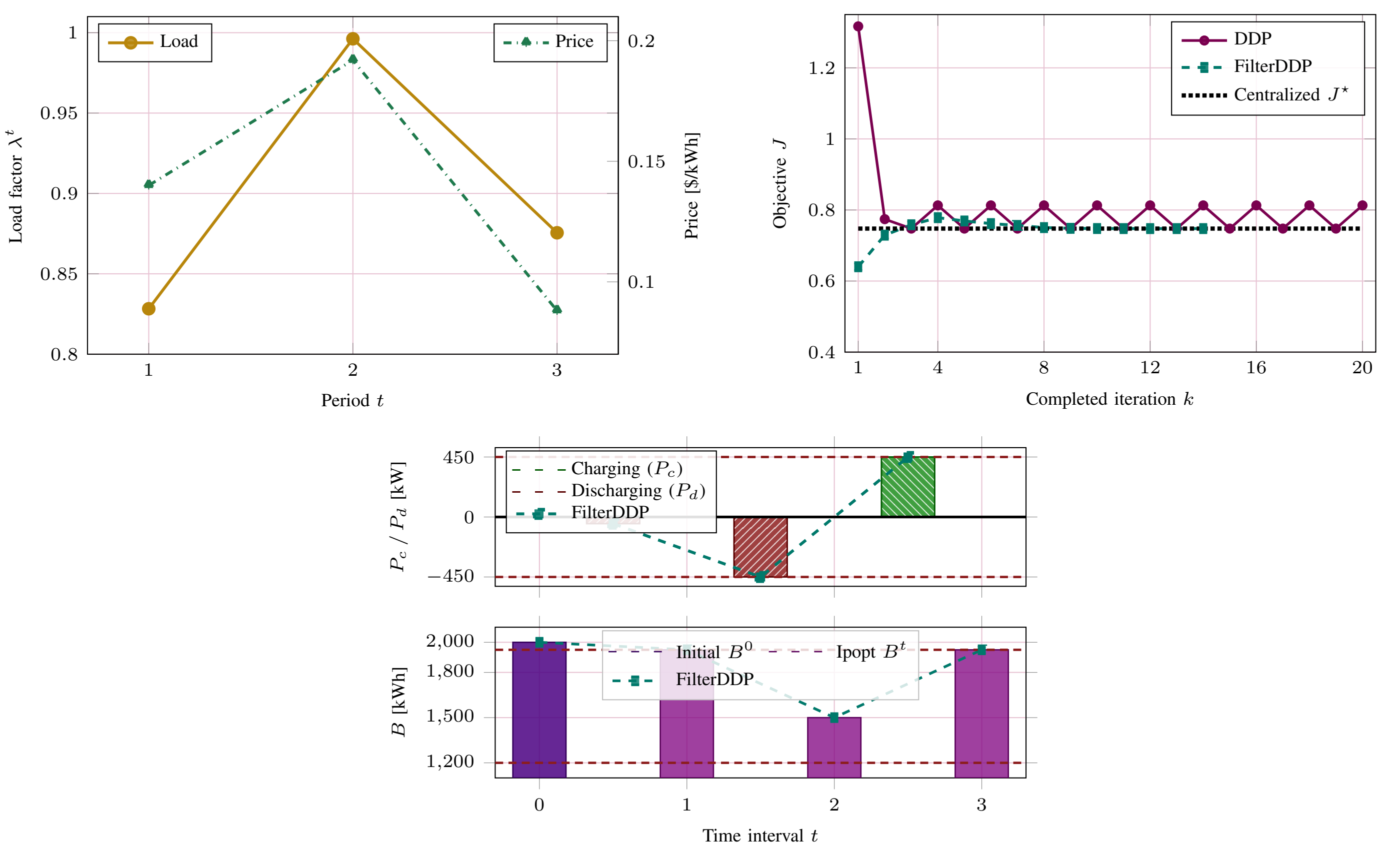

Fig. 2. Three-period inputs, objective trajectories, and optimal battery actions.

TABLE III
SOC EQUALITY MULTIPLIERS DURING THE TWO-CYCLE

k	$\lambda_{\text{soc}}^{k,1}$	$\lambda_{\text{soc}}^{k,2}$	$\lambda_{\text{soc}}^{k,3}$
15	0.135 00	0.088 04	0.314 29
16	0.088 04	0.231 96	0.088 04
17	0.135 00	0.088 04	0.314 29
18	0.088 04	0.231 96	0.088 04
19	0.135 00	0.088 04	0.314 29
20	0.088 04	0.231 96	0.088 04

TABLE IV
IDENTICAL-INSTANCE SOLVER COMPARISON

Method	Outcome	Iterations	Objective J
Centralized Ipopt	optimal	—	0.747650349
FilterDDP	optimal	14	0.747650349
First-order DDP	two-cycle	> 2000	0.747650353/0.812950270

The undamped sweep does not converge. Through 2000 sweeps it remains in an exact period-two cycle: the residual in (32) is 0.602079, and the objective alternates between 0.74765035 and 0.81295027. The centralized objective is 0.747650349; on the other branch the battery dispatch differs by as much as 0.850 p.u. (850 kW). Thus even this three-period copper-plate problem exposes the instability of the first-order iteration. On the identical data, bounds, and initialization, FilterDDP terminates normally in 14 iterations. Its objective differs from centralized Ipopt by 1.48×10^{-10}, while its maximum battery-power and energy differences are 6.50×10^{-9} p.u. and 3.71×10^{-9} p.u.h, respectively. Consequently, the two-cycle is a property of the first-order DDP iteration, not of an unusual or ill-posed optimization instance.

D. LinDistFlow Diagnostic Model

When a full branch-flow instance fails the centralized feasibility gate, the following LinDistFlow equations are used only to diagnose or initialize the network model:

$$\sum_{k \in \mathcal{C}(j)} P_{jk}^t - P_{ij}^t = P_{B,j}^t + p_{D,j}^t - p_{L,j}^t, \tag{33}$$

$$\sum_{k \in \mathcal{C}(j)} Q_{jk}^t - Q_{ij}^t = q_{D,j}^t - q_{L,j}^t, \tag{34}$$

$$v_j^t = v_i^t - 2\big(r_{ij} P_{ij}^t + x_{ij} Q_{ij}^t\big). \tag{35}$$

They omit the $r_{ij}\ell_{ij}^t$ and $x_{ij}\ell_{ij}^t$ loss terms, the quadratic current term in the voltage equation, and the SOC current constraint. All device, storage, substation, and voltage-bound constraints are otherwise unchanged. This approximation is not substituted for the loss-aware SOCP in reported OPF comparisons; it is identified explicitly whenever used as a feasibility diagnostic or starting-point construction.

REFERENCES

[1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE)*. IEEE, 2025, pp. 04–06.